

MDM & 8760 OVERHAUL

The aim of the project is to update the processes and tools used to generate one of the core Energy Production Estimate (EPE) deliverables and inform loss estimates with the NA Renewables Wind Energy Advisory Team

The project is divided into three parts, which can be completed sequentially or somewhat concurrently as the project progresses. It will involve manipulating meteorological data, using industry standard ULS-developed software, coding in Python and scaling for use in production line assessments.

At the end, we will have a more streamlined approach to generating select analysis components, and improved quality in time series analysis through integration of novel loss modeling techniques.

Note: Rev0 document is currently focused only on Part 1

Part 1: TSEC Mast File Monthly Scaling Tool

MILESTONES TO COMPLETION[BO1.1]

1. 2026-07-07: Get all scaling functionality working for Speed, Temperature, TI and Air Density
 - HOURS NEEDED ESTIMATE = 12-15?
2. 2026-07-13: Get monthly scaled CSV export working for 12-month Mast File, using the single or multiple 12-month mast file inputs
 - HOURS NEEDED ESTIMATE = 1
3. 2026-07-20: Get the monthly scaled CSV export working for 12-month Mast File(s), using the single or multiple MULTI-YEAR mast file inputs
 - HOURS NEEDED ESTIMATE = 1
4. 2026-07-27: Get WRF variables for pressure and humidity integrated and mast files exported in MM3.2 format using those
 - HOURS NEEDED ESTIMATE = 12-15??
5. 2026-08-03: Ensure working model in Mod1 and Mod4, using simple folder inputs and double-click applications for running
 - HOURS NEEDED ESTIMATE = 12-15

Part 2: Implement more complete loss modeling in OW & Automation

Part 3: Update Post Processing MDM Template for use in Automation and Standalone.

Meeting Notes

2026-07-24

Steps for the window selection and scaling application when >12 months in input target

- 1 – find overlapping period (when more than one mast)
- 2 – step through overlapping period in 12 month rolling windows, month by month
- 3 – find “best” 12 month window – min MAD
- 4 – store average temperature and speed from full input target files at each height (this will be used for final scale)
- 5 – trim targets to that 12 month window from step 3
- 6 – apply monthly scalings to trimmed targets from step 5 above
- 7 – apply final scaling to bring overall average of trimmed and monthly scaled targets back to speeds and temperatures from step 4

2026-07-20

For Wednesday:

Fix Air Density calc and elevation from pre-scaled temperature.

Get output working – format the header line by line to match the input OW format – Openwind 2.1 format

Code optimizing:

A lot of read/write – we should possibly re-optimize but not yet.

How to get it to run – command line currently – but we will run from a config

A lot of outputs – make a “working” folder for most

Analysts will want to see table of monthly scale values, and before and after speeds

2026-07-13

Separate files causing issues and filename convention

Precip and humidity working from wrf

2026-7-01

Beanan to add bulk scaling guidance

Grace to target completing all scalings recreating target mast file on simple case.

Moving any WRF parameters and MM3.2 (OW met mast format) to after that

2026-06-22

https://ul-my.sharepoint.com/:p/p/117068_global/IQDQJDQku3XTQa6a8TbR2GVYAZuRgAqzxjsO_XFhx38Vp-w?e=0U3Qzt

Didn't meet previous week –

Working on rolling averages, monthly profiles and scores

- Improve on error prone shear

Check temperature scaling

Clarify what to do with multi mast multi year

2026-06-15

Air Density Function in Python:

```
def rho(T_C, z, P0=101325, R=287, g=-9.8):  
    T_k = T_C + 273.15  
    exponent = (g*z*(1.0397 - (0.000025*z))) / (R*T_k)  
    rho = (P0*np.exp(exponent)) / (R*T_k)  
    return rho
```

Solve this for z

Example of it set up in Excel is here

Y:\ResearchandDevelopment\MDM-8760\TSEC_Mast_File_Tool\BaronWinds_InputData\00_Prep\AirDensity

It is sensitive to rounding – so best to calculate elevation per timestep (ignore any -999s) and then take the average of the time series of elevations.

Beanan – explicitly describe the TI

TI in the target mast file = Mean * SD

To scale for a new TI

Assume constant SD, so TI changes with mean speed only

- 1 - Calculate SD per time step before monthly scale is applied ($SD = TI / \text{Mean}$),
- 2 - Apply Monthly Scale
- 3 - Recalculate TI per time step from newly scaled mean speed from step 2 x SD from step 1

Example calculation ->

https://ul-my.sharepoint.com/personal/05597_global_ul_com/Documents/simpleTI-Calc_MastFile.xlsx?web=1

2026-06-12

- Grace to start looking at the rolling 12-month window while waiting on Mitchel and Beanan to confirm some items
- Mitchel / Beanan to confirm how to approach scaling to ensure overall average speed is maintained.

https://ul-my.sharepoint.com/personal/117068_global_ul_com/Documents/Temp%20Scaling.xlsx?web=1

- Start to work on timelines and milestones to keep on track

2026-06-10

- Get 2 target masts – done
- Prioritize get scaling working for speed and temperature first, export as appended to a header file
- Air Density – find how to adapt to the scaling

2026-06-08

Intro and KO following sharing of the report

Part 1 - Specification & Build Guide

Python-based monthly ratio scaling | seasonal correction

Draft for Internal Use

1. Purpose & Background

This document specifies a Python tool for adjusting Openwind Mast Files to represent long-term monthly profiles wind conditions at a site, for use in time series analysis (e.g. bat curtailment modeling) and products (12x24 aka MDM, and 8760). The method uses monthly ratio scaling for Wind Speed and Temperature, and any associated changes to Turbulence Intensity and Air Density.

The core idea is straightforward: a short target dataset (typically 1–3 years from an on-site mast) may have been collected during a period that was windier or calmer than average. By comparing the monthly profiles of the reference dataset that are concurrent to the target data, against the long term monthly profiles from the reference, 12 scaling factors are devised and applied to all project target mast files. In this way, each month in the target reflects the long-term profile. After the monthly correction, the overall average wind speed of the full 12 months is to be scaled back to the original annual speed —thereby preserving the target’s overall mean wind speed (which is assumed to already be determined in the WRA analysis). It will be a process similar to what is shown below from one of the Windographer functions below – but fully automated to optimize analyst time.

Note: This tool is designed to be used as part of a wider EPE workflow. Its outputs feed directly into energy yield calculations and are therefore subject to review. All scaling decisions must be logged and traceable.

1.1 Key Terminology

Term	Definition
------	------------

Target dataset	Openwind Mast Files -- Short-term wind measurement data from the site of interest (e.g. met mast or LiDAR). 10-minute or hourly resolution (e.g. WRF VMM), 100% data recovery, already scaled to long term speeds at “standard” hub heights (50/80/110/140).
----------------	--

Reference dataset	Long-term wind dataset used to derive the climatological profile. Typically reanalysis (ERA5, MERRA-2) or a WRF time series. Hourly or 10-min resolution.
-------------------	---

Monthly profile	The 12 mean wind speed values (one per calendar month) computed from a dataset over its full record.
-----------------	--

Scaling ratio	The multiplier applied to each 10-min value in a given calendar month. Ratio = reference long-term monthly mean / reference overlap monthly mean.
---------------	---

Overlap period	The period where target and reference data are both available. Used to compute monthly means for ratio calculation.
----------------	---

MAD score	Mean Absolute Deviation of the 12 scaling ratios from 1.0. A score of 0 means no correction needed; the target already matches the reference climatology perfectly.
-----------	---

Best window	The 12-month rolling window within the overlap that required the least correction, identified by lowest combined MAD score across all concurrent targets.
-------------	---

2. Input Data Specification

2.1 Target Dataset(s)

- 10-minute resolution time series
- 100% data recovery — no gap-filling is required or expected; this is handled in pre-processing before this tool is invoked
- One or more targets per project; all targets within a project must be exactly concurrent (identical start/end timestamps)
- Openwind MM 2.1 or MM 3.2 format – Output will be MM 3.2 to facilitate icing model
- Accepted formats: CSV

2.2 Reference Dataset

- Hourly or 10-minute resolution
- Must overlap with the target period; ideally extends significantly beyond it (10–20+ years recommended)
- Different structure from Target but will contain Speed and Temperature
- Multiple references can be provided for blending (see Section 5.8)

2.3 Project Configuration (adapt as necessary)

Option project run is defined by a configuration specifying:

- Project name / ID
- Folder containing Target Mast Files
- Folder containing Reference Datasets
- TBD if we need a configuration file that contains target HH speeds and dataset elevations (for Air Density)
- Output directory
- MAD combination method (mean, max, or weighted — see Section 5.5)

3. Method Overview

The pipeline processes each project in seven logical stages. All stages are deterministic — given the same inputs and config, outputs must be identical on every run.

Stage Description

1. Ingest & align Load target(s) and reference. Validate timestamps and resolution. Identify overlap period.
2. Reference LT profile Compute long-term monthly mean speeds from the full reference record (12 values). Blend if multiple references provided.
3. Reference overlap profile Compute monthly mean speeds from the reference data, restricted to the overlap period only.
4. Compute scaling ratios Divide reference LT profile by reference overlap profile to get 12 monthly scaling ratios. Report extreme ratios, possibly cap.
5. Rolling window search Slide a 12-month window across the overlap, and compute scaling ratios for each. Determine the “best” to use.
6. Apply scaling Apply ratios to: (a) best window — one corrected 12-month series per target; (b) each full calendar year in the target; (c) full target period if no complete calendar year exists, or there are multiple years.
7. Output Write adjusted 10-min series and diagnostic summary files to the output directory. — do this in MM 3.2 format

4. Detailed Step-by-Step Specification

4.1 Data Ingestion & Alignment

1. Load target(s) and reference into pandas DataFrames with a DatetimeIndex.
2. Validate that target resolution is 10-minute and reference is hourly or 10-minute.
3. Identify the overlap period: the intersection of the target date range and the reference date range. Raise an error if overlap is less than 12 months. Do not allow partial months, always start at the beginning of a month.
4. If multiple target masts are provided, find the common overlap window for these before comparing to the reference for overlap.

4.2 Long-Term Reference Monthly Profile

5. For each calendar month (1–12), compute the mean wind speed across the full reference record.
6. Result: a 12-element array `ref_lt_mean[month]`. This is the climatological target profile.
7. Store the reference profile in the output diagnostics regardless of blending.

4.3 Reference Overlap Monthly Profile

8. Restrict reference time series to the overlap period only.
9. For each calendar month (1–12), compute the mean wind speed.
10. Result per Reference: `ref_overlap_mean[month]` — 12 values.[GQ2.1]

4.4 Scaling Ratio Calculation[GQ3.1]

11. For each calendar month: $\text{scale}[\text{month}] = \text{ref_lt_mean}[\text{month}] / \text{ref_overlap_mean}[\text{month}]$
12. Apply the ratio multiplicatively to every 10-min timestep in the corresponding calendar month: $\text{adjusted_speed}[t] = \text{orig_speed}[t] \times \text{scale}[\text{month}(t)]$
13. After scaling, compute the new overall mean of the corrected series. Apply a single global factor to restore it to the original target mean. This preserves energy content while reshaping the seasonal profile. If the original target mean is not the intended target, provide targets to scale back to (TBD on this point, can get that information across easily)
 - Bulk Scaling Methodology:
 1. Calculate original speed and temperature averages for each height in the original target mast file (this is the single long term value to be corrected back to). Store.
 2. After applying ratios – calculate the new speed and temperature average in the adjusted target mast file.
 3. Scale the new overall average speed from the monthly adjusted target back to the original speed and temperature (for temp, consider K for the scaling)
 4. NOTE – if we are dealing with target masts with multiple years, but producing a 12-month monthly scaled mast file from that, then the average speed from the original target with all the years of data is the one stored from Step 1 above, and that is what the selected 12-month, monthly adjusted, target is scaled back to. [BO4.1]

15.14. Cap ratios option: if any ratio falls outside a configurable range (default: 0.70–1.30), apply the cap, log a warning, and flag that month in the diagnostics. A ratio outside this range likely indicates a physical mismatch between sites rather than a climatological signal.

4.5 Rolling 12-Month Window Search

16.15. Define windows using calendar month boundaries (e.g. Jan–Dec, Feb–Jan, Mar–Feb, etc.). This is the standard convention.

17.16. For a target of N months, there are N – 11 possible windows. For a 3-year (36-month) target: 25 windows.

18.17. Leap day handling: when enumerating window boundaries, prefer boundaries that avoid Feb 29. If a window start or end falls on Feb 29, shift it forward to Mar 1. Include Feb 29 only where unavoidable (e.g. a full calendar year 2024 output).

19.18. For each window, compute what the scaling ratios would be using only data from that window.

20.19. Score each window: $MAD = \text{mean}(|\text{scale}[\text{month}]| - 1.0)$ per target. Combine across targets using the configured method (default: mean MAD across all targets).

21.20. Rank all windows by combined MAD. Retain the full ranked table in the diagnostics output.

22.21. Select the window with the lowest combined MAD as the best window. This is one of the corrected outputs.

4.6 Applying Scaling & Producing Outputs

Using ratios derived for each window, produce the following corrected outputs per target:

- Best 12-month window
 - The 12-month slice identified in Stage 5, corrected using associated ratios, 12-month average speed corrected to LT speed (from original full MF or from input configuration sheet)
- Calendar year output(s)
 - One corrected series per full calendar year present in the target (Jan 1 00:00 to Dec 31 23:50).
 - Each year mean forced to same LT speed (from original full MF or from input configuration sheet)
 - If no full calendar year exists in the target, skip this output.
- Full-period output (off-calendar fallback and longer than 12-month time series)
 - If the target contains no full calendar year (e.g. March 2024 – February 2025), produce one corrected series for the entire target period.
 - Mean-preserved to the LT target mean

Note: A target may produce both calendar year outputs and a best window output. These are independent.

5. Design Considerations & Implementation Notes

5.1 Module Structure

Organise the codebase as follows:

Module	Responsibility
ingestion.py	Load CSVs/Parquet, validate timestamps and resolution, compute overlap period.
profiles.py	Compute monthly means from any time series. Blend multiple reference profiles.

scaling.py	Compute ratios, apply scaling, enforce caps, preserve mean.
search.py	Rolling 12-month window enumeration, MAD scoring, best window selection.
pipeline.py	Top-level orchestration: runs all stages for a given project config. Entry point.
output.py	Write corrected series and diagnostics to disk. Enforce naming convention.
viz.py	Diagnostic plots (monthly profile before/after, MAD score bar chart, ratio heatmap).
config.py	Load and validate JSON project config. Set defaults.

5.2 Output File Naming Convention

All output files follow a consistent naming pattern:

`{project}_{target_id}_{length_in_months}_{type}_TSEC_MF_Monthly_Scaled.csv`

Where `{type}` is one of: `best_window`, `cal_year_{YYYY}`, `full_period`.

Example: `BaronWinds_0246_12-Months_best_window_TSEC_MF_Monthly_Scaled.csv`

5.3 Diagnostics Summary Table

In addition to corrected time series, produce a single summary CSV per project containing:

- Window start and end dates
- Combined MAD score
- Per-reference scaling ratios for each of the 12 calendar months
- Flag indicating whether any ratio was capped
- Flag indicating whether the window contains a leap day

5.4 Computational Performance

- 10-min data for 3 years $\approx$ 157,000 rows per target. This is fast in NumPy/pandas — no optimisation is needed for typical project sizes.
- Pre-compute the reference long-term profile once per project and cache it. Do not recompute inside the window loop.
- Apply scaling using vectorised month-mapped array operations, not row-wise loops.
- For projects with many targets or very long records, consider Polars as a drop-in replacement for pandas — substantially faster for grouped aggregations.

5.6 Ratio Capping

Scaling ratios outside a physically plausible range introduce the potential for excessive scaling of the target at the 10-minute level, and potential distortion of plausible distributions. The default cap is 0.70–1.30 (i.e. $\pm 30\%$) for wind speeds, and 0.85 – 1.15 for temperatures. This is configurable.

- If a ratio is capped, the cap value is applied and a warning is logged.
- The diagnostics flag which months were capped and by how much.
- If more than 3 months in a single target are capped, consider raising this as a quality concern and deprioritize from best rolling window selections.

5.7 Leap Day Handling

- When enumerating 12-month rolling window boundaries, prefer boundaries that do not fall on Feb 29. Shift to Mar 1 if necessary.
- For full calendar year outputs (e.g. 2024), Feb 29 is unavoidably included. No special treatment — just document it in the output metadata.
- Do not let leap day avoidance override a meaningfully better MAD score. It is a tie-breaking preference only.
- Tag outputs that contain a leap day in the diagnostics summary, so downstream energy yield tools can handle 8,760 vs 8,784-hour years correctly.

5.8 Blended Reference Profile

When multiple reference datasets are provided, blend their long-term monthly ratios into a single 12-value ratios. Blending happens at the monthly profile level only — do not blend raw time series.

- Equal weight: median of all reference profiles (default).
- Correlation-weighted: weight each reference by its R^2 against the target during the overlap period. Compute R^2 from monthly means.
- User-specified: explicit weights in the config file, must sum to 1.0.

Note: Flag in the diagnostics/output when reference ratio profiles disagree significantly. A spread of more than 10% in any month across references may indicate complexity that should and should be reviewed. It should not stop a process, but may mean we would drop a reference dataset in a re-run

6. Output Summary

Output Format One per... Description

Adjusted 10-min Mast Files time series — best window CSV Target 12-month period with lowest combined MAD. overall mean speed preserved from original time series or input config file.

Adjusted 10-min Mast Files time series — calendar year CSV Target × year Each full Jan–Dec year in the target, independently adjusted, overall mean speed preserved from original time series or input config file.

Adjusted 10-min Mast Files time series — full period CSV Target Produced only when no full calendar year exists. Full target duration, corrected overall mean speed preserved from original time series or input config file.

Window summary table CSV Project All rolling windows ranked by combined MAD. Includes per-reference MAD and monthly ratios.

Reference profile CSV Project The 12 long-term monthly means used (or blended profile).

Scaling ratio matrix CSV Target 12 months × N windows. Useful for visualising how stable ratios are across different periods.

Diagnostic plots PNG Target Monthly profile before/after, MAD score bar chart, scaling ratio heatmap.

7. Open Items – to be assessed after core speed scaling complete

The following topics will be specified in a subsequent revision once example datasets are available.

7.1 Turbulence Intensity Scaling

In addition to mean wind speed, turbulence intensity (TI) has a monthly seasonal pattern. A similar ratio-scaling approach may be applied to TI, but the mechanics differ: TI is a derived quantity (standard deviation / mean) and must be handled carefully to avoid double-counting when mean speed is also scaled. We will calculate a SD in each timestep from unadjusted speed * TI, then divide that SD by adjusted speed to get the new TI

TI in the target mast file = Mean * SD

To scale for a new TI

Assume constant SD, so TI changes with mean speed only

- 1 - Calculate SD per time step before monthly scale is applied ($SD = TI / \text{Mean}$),
- 2 - Apply Monthly Scale
- 3 - Recalculate TI per time step from newly scaled mean speed from step 2 x SD from step 1

7.2 Temperature Scaling

Air density correction requires accurate long-term temperature data. If the reference dataset includes temperature, a parallel monthly profile scaling can be applied to the target temperature record. A similar process to speeds is applied, but all temperature data is first converted from C to K to be scaled appropriately. A delta adjustment approach, e.g. additive for difference rather than scaling may also be explored.

UPDATE[BO5.1] 2026-10-06: Assume a scale (not an offset) – convert the reference and target temperatures from Celsius to Kelvin, $K = C + 273.15$.

Then perform the same scaling process to the temperatures (always working in Kelvin)

Then convert back to Celsius from Kelvin, $C = K - 273.15$

Note: Temperatures should always be in C in our standard reference datasets, but in case there is a K, maybe it's worth running a quick check on the average temperature of the reference to make sure it less than 200. If it's not, then maybe it doesn't need the Kelvin conversion.

This also affects air density in each timestep, we will need to figure out how to adjust the air density.

the formula to back calculate the elevation from an air density and temperature is here

"Y:\ResearchandDevelopment\MDM-8760\TSEC_Mast_File_Tool\BaronWinds_InputData\00_Prep\AirDensity\BaronWinds_Air_Density_Mast_0246_2025-06-18_FNL-FormulaCheck.xlsm"

- given the rounding of the temperatures, the elevations are error prone. we should calculate for all timesteps and then get the average elevation from that.

- we will have air density from different heights in the target mast files (e.g. 50m, 80m, 110m, 140m), so we should either
 - o 1 - calculate the elevation for each height (there will be a temperature and AD for each height) or
 - o 2 - calculate the elevation for a single height and apply the height differential to each of the other target mast file heights. e.g. calculate it for 50m, but then add 30m for 80, add 60m for 110, add 90m for 140m. So, if you got an elevation of 500m for the 50m height in the mast file, you'd then give a height of 530m, 560m, 590m to the air density calculation for the other heights.

7.2 WRF Icing Model Variables – Humidity and Precipitation

The MM 3.2 format carries humidity data per height, but we can assume one height representative of all, and precipitation data. We will try to find a way to get this data from a WRF time series into the MFs. We will assume that all projects will have long term hourly WRF dataset to facilitate this.

7.3 Summary Statistics comparison output

With any scaling of wind data from originally measured profiles, there is a risk that the resulting outputs still follow the overall mean speed of the original dataset but become skewed in distributions or wind roses, summary comparison graphs will be useful to quickly compare results, e.g. wind roses, speed distributions

7.4 Directional Correction

The current specification corrects omnidirectional wind speed only. A future extension may apply sector-wise corrections if both target and reference carry wind direction. This requires a different ratio matrix (12 months $\times$ N sectors) and significantly more overlap data to be statistically robust.

8. Suggested Getting-Started Sequence

Review and consider these steps, but they are not set in stone and Mitchel would be best to review ideas. Could also see what copilot might suggest.

23.22. Read this document in full. Note any questions.

24.23. Set up a Python environment with Mitchel's help: Chatbot suggests Python 3.10+, pandas, numpy, matplotlib, PyYAML.

25.24. Create the module structure (Section 5.1) with empty files and docstrings only. No logic yet.

26.25. Implement profiles.py: given a DataFrame with a DatetimeIndex and a speed column, return a 12-element monthly mean array.

27.26. Compute ratios in from reference datasets

28.27. Implement scaling.py: given a single target 12-month Baron Winds Mast File series and a 12-element ratio array, return the corrected series with mean preserved. Test on a Baron Winds datasets and verify the output mean matches the input mean.

29.28. Implement search.py: enumerate 12-month windows, compute MAD per window, return a ranked DataFrame. Test on a 36-month synthetic target.

30.29. Implement ingestion.py and pipeline.py to wire everything together.

31.30. Run on example Baron Winds datasets (2 x input MFs with different lengths of data) and review diagnostic plots.

9. Questions & Contact

Bring questions regularly, we will set up a chat to discuss and track progress. We'll have regular check ins with Beanan and Mitchel, If you find an ambiguity or inconsistency in this document, flag it — do not make an assumption and build on it.